VeriSight

AI-Powered Visual Verification System for Delivery Platform Fraud Detection

— Hack-Nocturne '26 | Cybersecurity Domain | Team: return0 —

PROJECT AT A GLANCE

Domain: Cybersecurity | **Focus:** Visual Image Fraud Detection | **Team:** return0

Tech: CNN | Vision Transformers | GAN Detection | OCR | FastAPI | React

— Hack-Nocturne 2026 —

1. Executive Summary

E-commerce, quick commerce, and food delivery platforms across India and the world process millions of refund and complaint requests every day. The vast majority of these claims rely entirely on user-uploaded product images — photographs submitted by customers as visual proof that something went wrong with their order.

This trust-based system is being systematically exploited. Customers are using freely available AI image editing tools, generative AI platforms, and Photoshop to digitally manipulate product images before submitting them as complaint evidence. The most common manipulation targets are product expiry dates — altered to make a fresh product appear expired — and packaging labels, which are digitally modified to create false impressions of damage, wrong contents, or contamination.

> *"Returns fraud and abuse accounted for 13.7% of all returns in 2023, amounting to over $100 billion in losses globally." — Loop Returns, 2024*

VeriSight is an AI-powered visual verification system purpose-built to detect these image manipulations in real time. It combines CNN-based image forensics, Transformer-based AI-generated image detection, GAN artifact analysis, and OCR-assisted text localization to evaluate every complaint image and produce an Authenticity Confidence Score — a single, explainable metric that tells the platform whether a submitted image is genuine or manipulated.

VeriSight is designed as a cloud-ready, API-first microservice that can be directly integrated into the complaint and refund workflows of any delivery-based platform — Zomato, Swiggy, Blinkit, BigBasket, Amazon, Flipkart, and beyond. It is practical, scalable, and deployable — not just a research prototype.

2. Problem Statement

2.1 The Refund Complaint Ecosystem

Modern delivery platforms have built their customer trust on generous and fast refund policies. When a customer claims they received an expired product, wrong item, damaged packaging, or contaminated food, platforms typically process the refund with minimal verification — relying almost entirely on the image submitted by the customer as proof.

This trust-first approach, while excellent for customer experience, has created a massive and growing attack surface. Fraudsters have discovered that submitting a manipulated image is the single easiest way to obtain an unauthorized refund — and with AI tools now freely accessible to anyone with a smartphone, image manipulation has become trivially easy.

2.2 The Core Fraud Patterns VeriSight Targets

Fraud Type	Detailed Description
Expiry Date Manipulation	Customer digitally alters the printed expiry date on a product's packaging — changing a valid future date to a past date — to falsely claim the delivered product was expired at the time of delivery
Packaging Label Manipulation	Brand name, product name, batch number, MRP, weight, or ingredient list on the packaging label is digitally edited using AI tools to create the appearance of wrong or counterfeit product delivery
AI Image Generation	Customer uses generative AI (Stable Diffusion, DALL-E, Adobe Firefly, Midjourney) to create an entirely synthetic product complaint image that never existed in reality
GAN-Manipulated Images	Generative Adversarial Network tools are used to realistically alter specific regions of a product photo — inserting damage, mold, incorrect labeling, or wrong packaging — while keeping the rest of the image looking photorealistic
Image Splicing / Compositing	Content from a different product image is composited into the complaint photo — for example, a damaged portion of another product's packaging is pasted over the delivered product's intact packaging
Recycled Complaint Images	Previously used complaint images (genuine or fake) are resubmitted for new orders across the same or multiple accounts to claim repeated refunds without submitting new manipulated images

2.3 Why Existing Systems Fail

- Platforms rely on human reviewers who cannot detect AI-generated or subtly edited images with the naked eye
- No automated system exists that specifically verifies the authenticity of complaint product images
- Manual review is slow (24–48 hours), inconsistent across reviewers, and completely unscalable at platform volumes

- Current rule-based fraud filters (e.g., flag if same user complains 3 times) do not examine image authenticity at all
- Generative AI tools produce images with photorealistic quality that are indistinguishable from real photos without algorithmic analysis
- Platforms that do have image checks typically use simple hash-based duplicate detection — easily bypassed by applying minor edits

2.4 Market Scale & Financial Impact

Metric	Data
Global e-commerce fraud losses (2024)	USD 48 billion annually — businesses lose an average of USD 207 for every USD 100 in fraud when factoring in chargebacks, shipping, and processing costs
Return & refund fraud globally (2024)	USD 103 billion. Returns fraud accounted for 13.7% of all returns in 2023. Abusive returns soared 64% between January 2024 and May 2025
India deepfake shopping scams (2024)	45% of Indian consumers or someone they know fell victim to deepfake shopping scams (McAfee Global Festive Shopping Survey 2024)
India example: Meesho fraud	A single organized group in Surat defrauded Meesho of Rs 5.5 crore through coordinated refund fraud — illustrating the organized scale of the problem
India consumer complaints growth	7,482 consumer grievances filed against food delivery apps in 2024–25 alone — a 10x increase from 805 in 2020–21

3. VeriSight — Solution Overview

VeriSight is an AI-powered visual verification system that automatically analyzes every user-submitted complaint image and produces an Authenticity Confidence Score between 0 and 100, along with a human-readable explanation of the factors that contributed to the score. The system is built around four core analysis modules that work in parallel, each targeting a distinct manipulation vector.

VeriSight Core Philosophy

VeriSight does NOT make the refund decision — it provides an evidence-based Authenticity Confidence Score with explainable insights that enables the platform to make a faster, fairer, and more accurate decision. The system supports human reviewers rather than replacing them.

4. Core Detection Modules

Module 1: CNN-Based Image Forensics

Purpose

A deep Convolutional Neural Network forms the foundational analysis layer of VeriSight. It performs pixel-level analysis of the submitted complaint image to detect signs of manual editing, region-level manipulation, and image compositing.

What It Detects

- Copy-move forgery — areas of the image that have been duplicated and repositioned (e.g., copying a damage region from elsewhere in the image or from another photo)
- Image splicing — seamless insertion of content from a different image source, detectable through pixel-level statistical inconsistencies at boundary regions
- Error Level Analysis (ELA) — JPEG images that have been edited show inconsistent compression artifact levels across regions; ELA maps reveal these as visually distinct areas of higher error levels
- Lighting and shadow inconsistencies — regions that have been digitally inserted typically have lighting angles and shadow directions that don't match the rest of the image
- Noise pattern analysis — genuine images have consistent sensor noise patterns; inserted regions from different sources show different noise profiles

Model Architecture

Component	Details
Primary Backbone	EfficientNet-B4 pre-trained on ImageNet — chosen for superior accuracy-to-computation ratio over ResNet-152 and VGG-19. 7.6x smaller and 5.7x

	faster than ResNet-152 with higher accuracy on forgery benchmarks
Forgery Localization	GCA-Net (Gated Context Attention Network) for pixel-level manipulation region segmentation — produces visual heatmaps showing exactly which pixels were manipulated
ELA Pipeline	Custom ELA-CNN pipeline that generates Error Level Analysis maps and feeds them as additional input channels alongside the RGB image to the classification model
Training Dataset	CASIA 2.0 (12,614 images), CoMoFoD (copy-move forgery), MICC-F220 (forgery localization ground truth), DF2023 (1 million+ images for robust generalization)
Benchmark Accuracy	ResNet-101 baseline achieves 93.46% accuracy on CASIA 2.0 (IEEE 2024). VeriSight targets >95% using EfficientNet-B4 + ELA pipeline combination

Module 2: Transformer-Based AI-Generated Image Detection

Purpose

This module specifically targets complaint images that were not edited from a real photo but were entirely or partially generated using modern generative AI tools — including Stable Diffusion, DALL-E 3, Midjourney, Adobe Firefly, and GAN-based image generators.

Why Transformers for This Task

Vision Transformers (ViTs) use self-attention mechanisms to model relationships between non-adjacent image patches — enabling them to detect long-range semantic inconsistencies that CNNs miss. Research shows ViTs achieve up to 96% accuracy on AI-generated image detection tasks. Crucially, transformers generalize better to unseen generative models than CNN-only approaches, making them ideal as new AI tools emerge.

Detection Mechanisms

- Semantic inconsistency detection — AI-generated images of products often contain logically inconsistent text, labels, or barcodes (e.g., partially hallucinated expiry dates, wrong brand fonts, blurry or garbled packaging text)
- Texture artifact analysis — GAN and diffusion model generated images often produce overly smooth textures or unnatural repetitive patterns in background regions
- Frequency domain analysis — AI-generated images have characteristic spectral signatures in the frequency domain (detectable via Discrete Fourier Transform and Discrete Wavelet Transform) that differ from real camera-captured images
- Cross-patch attention — ViT self-attention heads identify regions where generated content does not semantically belong in the scene context
- CLIP-based feature analysis — CLIP embeddings (vision-language model) inherently distinguish between AI-generated and real images at the representational level; CLIP-ResNet50 achieves 99.4% accuracy on StyleGAN detection (CVPR 2023)

Model Stack

Model	Role & Benchmark
Vision Transformer (ViT-B/16)	Primary AI-generated image classifier. ViTs achieve up to 96% accuracy using self-attention mechanisms to detect inter-patch semantic inconsistencies (imrjr.com 2025)
CLIP-ResNet50	GAN fingerprint detector. Achieves 99.4% and 99.2% accuracy on StyleGAN and StyleGAN2 generated images respectively (CVPR 2023, LGrad framework)
FatFormer (CLIP + FAA)	Generalizable synthetic image detection integrating Forgery-Aware Adapter and Language-Guided Alignment — effective on unseen generative models including diffusion models
Grad-CAM Explainability	Gradient Class Activation Mapping generates visual heatmaps of which image regions triggered the AI-detection decision — essential for human reviewer understanding

Module 3: GAN Artifact & Texture Inconsistency Analysis

Purpose

Even when a complaint image is not fully AI-generated, GAN-based inpainting tools can be used to realistically alter specific regions — replacing a genuine expiry date with a manipulated one, or inserting realistic-looking damage or mold into a fresh product photo. This module specifically targets these localized GAN-based edits.

GAN Artifacts Detected

- Spectral replication artifacts — GAN upsampling components introduce characteristic frequency-domain artifacts that appear as replications in the Fourier spectrum, detectable using AutoGAN-style spectrum classifiers (IEEE 2019)
- Checkerboard artifacts — transposed convolution upsampling in GAN generators produces characteristic checkerboard patterns in synthesized regions, visible in frequency analysis
- Color channel statistical discrepancies — GAN-generated regions often have subtle color channel correlations that differ from surrounding genuine image regions, detectable through per-channel statistical analysis
- Boundary blending inconsistencies — even sophisticated GAN inpainting leaves statistical traces at the boundary between the genuine and generated region
- Texture uniformity anomalies — GAN-generated text and numbers on product packaging show characteristic smoothness and regularity that differs from the natural variation of physically printed text

Detection Approach

VeriSight implements AutoGAN-style spectrum-based classification (IEEE 2019) combined with GAN re-synthesis detection (comparing the input image with its re-synthesized version across super-resolution, denoising, and colorization tasks). Discrepancies between the original and re-synthesized versions reveal GAN-inserted regions.

Module 4: OCR-Assisted Text & Expiry Date Verification

Purpose

This is VeriSight's most distinctive and unique module — one that no existing general-purpose image forensics system implements for the refund fraud use case. It uses OCR (Optical Character Recognition) to extract and analyze text content from product packaging in complaint images, specifically to detect digitally altered expiry dates, batch numbers, and label text.

Critical Design Principle

> *OCR plays a supporting role in VeriSight — it assists verification but is not used alone for decision-making. It is combined with image forensics evidence from Modules 1–3 before contributing to the final score.*

OCR Pipeline

1. Image preprocessing — grayscale conversion, CLAHE contrast enhancement, adaptive thresholding, noise reduction to maximize OCR accuracy on varied packaging surfaces and lighting conditions
2. Text region detection — YOLOv8 or DBNet (improved with CBAM attention module) localizes all text regions in the image, including dot-matrix printed expiry dates, laser-etched codes, and standard label text
3. Text recognition — EasyOCR (CRNN + CRAFT architecture) or PaddleOCR extracts the text content from each detected region; handles dot-matrix fonts, varied date formats (DD/MM/YY, MM-YYYY, DD.MM.YYYY, etc.), and multiple scripts
4. Date extraction & validation — regex pattern matching extracts all date-format strings and identifies the most likely expiry date candidate from the recognized text
5. Statistical plausibility analysis — the extracted date is analyzed for statistical plausibility against the platform's order date and delivery timestamp; dates in the distant past or implausibly short shelf lives are flagged
6. Text texture forensics — the visual texture of the extracted text region is analyzed by the CNN to detect whether the text was digitally inserted (smooth, pixel-perfect edges) vs. physically printed (natural ink variation, slight embossing texture)

OCR Technology Comparison

OCR Engine	Architecture	VeriSight Usage
EasyOCR	CRNN + CRAFT	Primary OCR engine — excellent adaptability to diverse layouts and packaging types; supports 80+ languages
PaddleOCR	PP-OCR v3	Fallback OCR for dot-matrix and degraded text; fine-tuned on packaging images
DBNet + CBAM	Detection Network	Text region detection and localization — improved backbone for complex package image backgrounds (Tandfonline 2023)
Tesseract v5 + LSTM	Classical + LSTM	Verification check for standard printed text — lower accuracy on dot-matrix but useful for cross-validation

Expiry Date Detection Research Basis

- DBNet + CBAM for expiry date detection achieves high accuracy and robustness on complex food package images (Tandfonline 2023)
- EasyOCR + CRNN with fine-tuning achieves strong performance on product expiry date recognition (ResearchGate 2020)
- ResNet-50 + VGG-16 regression model pipeline: ResNet-50 classifies product type, VGG-16 produces bounding boxes for the expiry date region, then OCR extracts the text (GitHub / Shivaneej 2022)
- R-CNN based character recognition achieves 91.1% precision and 80.9% F1 on pharmaceutical packaging text without handcrafted features (ResearchGate 2024)
- A generalized FCN-based framework for expiry date recognition handles varied fonts, formats, and packaging materials (ScienceDirect / Expert Systems with Applications 2022)

5. Authenticity Confidence Score Engine

The outputs of all four modules are synthesized by the Authenticity Confidence Score Engine into a single score from 0 to 100, where 100 means the image appears completely genuine and 0 means it is almost certainly manipulated. Each module contributes a weighted sub-score:

Module	Weight	Signal Provided
CNN Image Forensics	35%	Forgery probability: ELA map, copy-move score, splicing score, boundary analysis
Transformer AI Detection	30%	AI-generation probability: ViT attention score, CLIP embedding distance, frequency spectrum
GAN Artifact Analysis	20%	GAN manipulation probability: spectral replication, texture anomaly, boundary blending score
OCR Text Verification	15%	Text plausibility: date validity, font texture forensics, text region CNN analysis

5.1 Decision Thresholds

Score	Classification	Platform Action	Notes
85–100	Genuine	Auto-approve refund	Image verified as authentic; refund processed automatically within 60 seconds
60–84	Likely Genuine	Fast-track review	High probability genuine; human reviewer spot-checks within 2 hours
35–59	Suspicious	Manual review + proof request	Refund held; customer asked for additional video evidence or original packaging photo
0–34	Manipulated	Reject + flag account	Refund denied; account flagged; repeat fraudsters permanently banned; evidence report generated

5.2 Explainable AI (XAI)

Every VeriSight decision is fully explainable. The system generates a human-readable evidence report for each complaint evaluation, containing:

- Overall Authenticity Confidence Score with breakdown by module
- Visual heatmap overlay on the complaint image showing suspected manipulation regions (from GCA-Net + Grad-CAM)
- SHAP-based feature importance explanation showing which signals drove the score
- OCR-extracted text with date plausibility analysis result

- Recommended action with confidence level

This explainability is critical — it enables human reviewers to audit every decision, understand edge cases, and override the system when needed. It also makes VeriSight's outputs usable as legal evidence.

6. System Architecture

VeriSight is built as a cloud-ready, API-first microservice system. It is designed to integrate directly into the existing complaint and refund workflows of any delivery platform via REST API, with a target end-to-end processing latency of under 5 seconds per image.

6.1 Processing Pipeline

7. Customer submits complaint with product image via platform app
8. Platform sends image + complaint metadata to VeriSight REST API
9. API Gateway validates the request and queues the image for processing
10. Image is routed simultaneously to all four analysis modules (parallel processing)
11. Module 1 (CNN) returns: forgery probability + ELA heatmap + GCA-Net manipulation map
12. Module 2 (Transformer) returns: AI-generation probability + Grad-CAM heatmap + CLIP score
13. Module 3 (GAN) returns: GAN artifact probability + spectral analysis result + boundary score
14. Module 4 (OCR) returns: extracted text, date plausibility flag, text texture forensics score
15. Score Engine aggregates all four outputs into Authenticity Confidence Score (0–100)
16. SHAP + Grad-CAM generate explainability report with visual heatmap overlay
17. Final response returned to platform API: score + decision + explanation + heatmap image
18. Platform updates complaint status and takes appropriate action

6.2 Architecture Layers

Layer	Component	Technology
API Gateway	Complaint intake endpoint	FastAPI (async) + Pydantic validation + JWT authentication
Parallel Processing	Module orchestration	Celery distributed task queue + Redis broker for concurrent module execution
CNN Module	Image forensics inference	PyTorch + EfficientNet-B4 + ONNX Runtime (fast inference) + OpenCV for ELA
Transformer Module	AI-image detection inference	PyTorch + ViT-B/16 + HuggingFace Transformers + CLIP (OpenAI)
GAN Module	GAN artifact analysis	NumPy + SciPy (FFT spectrum analysis) + custom AutoGAN spectrum classifier
OCR Module	Text extraction pipeline	EasyOCR + PaddleOCR + YOLOv8 text detection + Regex date parsing
Score Engine	Score aggregation + XAI	Python + SHAP + Grad-CAM visualization + PIL for heatmap overlay
Database	Complaint & result storage	PostgreSQL (complaint history, audit log) + Redis (feature cache, model cache)
Frontend Dashboard	Admin monitoring UI	React.js + Recharts (analytics) + Leaflet.js

		(geographic visualization)
Infrastructure	Cloud deployment	Docker + Kubernetes + AWS EC2 / GCP + GitHub Actions (CI/CD)

7. Full Technology Stack

7.1 Frontend & Backend

Technology	Purpose & Justification
React.js	Admin monitoring dashboard — real-time fraud score feeds, complaint review queue, analytics charts, and heatmap visualization
Python 3.11	Primary backend language — extensive ML/AI library ecosystem (PyTorch, Scikit-learn, OpenCV, EasyOCR) makes Python the only practical choice for this system
FastAPI	High-performance async REST API framework — async request handling enables parallel module processing without blocking; automatic OpenAPI documentation for easy platform integration
Celery + Redis	Distributed async task queue — enables all four VeriSight modules to run in parallel rather than sequentially, achieving the target sub-5-second end-to-end latency
PostgreSQL	Primary relational database for storing complaint records, VeriSight score history, fraud flags, account-level analytics, and full audit trail

7.2 AI & ML Stack

Technology	Purpose & Justification
PyTorch	Primary deep learning framework for training and serving all CNN, ViT, and GAN detection models. Dynamic computation graph enables flexible model architecture experimentation
EfficientNet-B4	CNN backbone for image forensics module — state-of-the-art accuracy-to-parameter ratio. Pre-trained on ImageNet, fine-tuned on CASIA 2.0 forgery dataset
Vision Transformer (ViT)	AI-generated image detector — self-attention mechanisms detect cross-patch semantic inconsistencies; achieves 96% accuracy on synthetic image detection tasks
CLIP (OpenAI)	Vision-language model for GAN fingerprinting — CLIP-ResNet50 achieves 99.4% accuracy on StyleGAN/StyleGAN2 detection (CVPR 2023)
GCA-Net	Pixel-level forgery localization using gated context attention — produces interpretable manipulation heatmaps for human reviewer display
EasyOCR + PaddleOCR	Text extraction from product packaging — EasyOCR handles varied fonts and layouts; PaddleOCR handles dot-matrix expiry date printing
YOLOv8 / DBNet+CBAM	Text region detection on product packaging — localizes expiry date fields, label text, and barcode regions for targeted OCR extraction
SHAP + Grad-CAM	Explainable AI — SHAP provides feature-level score attribution; Grad-CAM generates visual activation maps showing which image regions

	drove the model decision
ONNX Runtime	Optimized model inference runtime for production deployment — converts PyTorch models to ONNX for platform-independent, low-latency inference
OpenCV + Pillow	Image preprocessing (grayscale, thresholding, noise reduction), ELA map generation, heatmap overlay rendering on complaint images

8. Dataset Strategy

VeriSight requires training data across two distinct domains: general image forgery detection (using existing public benchmarks) and product-specific packaging/expiry date manipulation (custom dataset to be built). The team has a concrete three-pronged strategy.

8.1 Public Benchmark Datasets (Image Forgery)

Dataset	Size	Use in VeriSight
CASIA 2.0	12,614 images	Primary CNN training dataset for splicing and copy-move forgery detection
CoMoFoD	260 image sets	Copy-move forgery with post-processing (blur, noise, compression) robustness testing
DF2023	1M+ images	Large-scale training for model generalization; covers diverse forgery types
MICC-F220	220 images	Pixel-level forgery localization ground truth for GCA-Net heatmap validation
ImagiNet (200K images)	200,000 images	Real vs AI-generated image classification training for ViT module (Boychev et al. 2024)
FaceForensics++	1,000 videos	Deepfake / GAN detection pre-training for the AI-image detection module

8.2 Custom VeriSight Product Dataset (Built from Scratch)

Since no public dataset exists specifically for product packaging image manipulation in the refund fraud context, the team will construct a custom dataset:

19. Photograph 150+ real product packages from categories: snacks, beverages, dairy, medicines, cosmetics, grocery
20. Create genuine complaint versions: photograph packages at slightly poor angles, under varying lighting
21. Create expiry-date-manipulated versions: digitally alter dates using Photoshop, Canva, and AI inpainting tools
22. Create label-manipulated versions: alter brand names, weights, MRPs, batch numbers on packaging
23. Create AI-generated versions: generate synthetic product complaint images using Stable Diffusion, DALL-E 3, and Adobe Firefly
24. Apply augmentation: rotation, brightness variation, JPEG re-compression at varied quality levels, noise addition
25. Label dataset: manipulation type, manipulated region pixel mask, manipulation tool used, confidence level

Target: 4,000+ labeled product complaint images across 5 classes (genuine, expiry-manipulated, label-manipulated, AI-generated, GAN-inpainted).

9. Research Foundation

VeriSight is grounded in 25+ peer-reviewed research papers spanning image forensics, OCR, GAN detection, transformer models, and e-commerce fraud. The following tables summarize the key papers by module.

9.1 Image Forgery & CNN Detection

#	Paper	Source / Year	VeriSight Application
1	Image Forgery Detection: Survey of Recent Deep Learning Approaches	Multimedia Tools & Applications, Springer 2022	Architecture selection foundation for CNN forensics module
2	Detection of Image Forgery using ELA-CNN, VGG19, EfficientNet-B2	AIP Conference 2024	ELA-CNN pipeline design; direct benchmark comparison for Module 1
3	GCA-Net: Gated Context Attention for Image Forgery Localization	CVPR Workshops, IEEE 2022	Pixel-level manipulation heatmap architecture
4	HiFi-Net++: Language-Guided Hierarchical Fine-Grained Forgery Detection	IJCV, Springer 2024	Multi-level forgery detection for product image analysis
5	Exploring ML Approaches for Efficient Image Forgery Detection (98.5% accuracy)	J. Forensic Sciences, Wiley 2025	State-of-art accuracy benchmark target for VeriSight CNN module
6	DF2023: Digital Forensics Dataset for Image Forgery Detection	arXiv 2025	Large-scale training data source for generalization

9.2 GAN & AI-Generated Image Detection

#	Paper	Source / Year	VeriSight Application
7	Learning on Gradients: Generalized Artifacts Representation for GAN-Generated Images (LGrad)	CVPR 2023 / IEEE	GAN artifact detection via gradient-based generalized representation
8	Detecting and Simulating Artifacts in GAN Fake Images (AutoGAN, spectrum classifier)	IEEE WIFS 2019	Spectrum-based GAN artifact classification — Module 3 core approach
9	Methods and Trends in Detecting Generated Images: Comprehensive Review	arXiv 2025	Survey covering GAN, diffusion, VAE detection methods — architecture selection reference
10	Advanced Detection of AI-Generated Images Through Vision Transformers	TechRxiv 2024	ViT-based synthetic image detection — Module 2 architecture reference
1	Breaking Semantic Artifacts for	NeurIPS 2024	Generalizable AI image detection

1	Generalized AI-Generated Image Detection		framework applicable to new generative models
12	FatFormer: Forgery-Aware Adapter + Language-Guided Alignment for Synthetic Detection	CVPR / arXiv 2024	CLIP-based generalizable synthetic image detector — key component of Module 2
13	Vision Transformers & Statistical Methods in Deepfake Detection (96% accuracy)	IMRJR 2025	Hybrid ViT + statistical artifact approach for robust detection
14	AI vs. AI: Can AI Detect AI-Generated Images? (GAN classification study)	PMC / MDPI 2023	Validation of AI-detection approach and failure case analysis for edge cases

9.3 OCR & Expiry Date Detection

#	Paper	Source / Year	VeriSight Application
15	Recognition of Expiry Data on Food Packages Based on Improved DBNet (with CBAM)	Tandfonline 2023	DBNet+CBAM architecture for text detection on complex packaging — core OCR Module design
16	A Generalized Framework for Recognition of Expiration Dates Using Fully Convolutional Networks	ScienceDirect / Expert Systems 2022	FCN-based framework handling varied fonts, formats, and packaging types
17	Expiry Date Recognition Using Deep Neural Networks (EasyOCR + CRNN + CRAFT)	ResearchGate 2020	EasyOCR fine-tuning approach for product expiry date extraction
18	ML-Enabled Expiry Date Recognition and Object Classification of Grocery Products (ResNet-50 + VGG-16 + OCR)	GitHub / Shivaneej 2022	Two-stage pipeline: product classification then expiry date extraction
19	Integration of LLM in Expiration Date Scanning (R-CNN: 91.1% precision, 80.9% F1)	ResearchGate / SCPE 2024	R-CNN benchmark for character recognition on pharmaceutical packaging
20	AI and OCR Combine for Expiration Date Inspection (Neurala VIA + OCR pipeline)	Vision Systems Design	Industry validation: AI + OCR combination outperforms each technology alone
21	A Mobile Approach to Food Expiration Date Determination Using OCR and Firebase ML	MDPI 2025	Mobile-first OCR pipeline reference for real-world deployment considerations
22	A Comparative Study on Automated Expiry Date Extraction from Official Documents	IARIA AISyS 2025	OCR engine comparison (EasyOCR vs PaddleOCR vs Tesseract) for VeriSight engine selection

10. Cybersecurity Domain Justification

Cybersecurity Sub-Domain	How VeriSight Addresses It
Digital Image Forensics	CNN-based detection of image manipulation, ELA forensic analysis, and pixel-level forgery localization — core digital forensics methodology applied to complaint images
Deepfake & Synthetic Media Detection	ViT + CLIP + GAN spectrum analysis specifically targeting AI-generated product images — addressing one of the fastest-growing cybersecurity threats
Financial Fraud Prevention	Protecting platform revenue from organized refund fraud — a financial cybercrime costing the industry USD 103 billion annually in fraudulent returns
AI Misuse Detection	Detecting the weaponization of generative AI tools (Stable Diffusion, DALL-E, Firefly) against commercial platforms — a specifically cybersecurity-relevant AI safety concern
Digital Evidence & Forensic Reporting	Generating SHAP-explainable, legally defensible evidence reports of detected image manipulation — applied digital forensics for consumer fraud investigation
Authenticity Verification Systems	Building a scalable system for verifying the integrity and authenticity of digital visual evidence — a core cybersecurity trust infrastructure problem

11. Hackathon Development Roadmap

Phase 1 — Core Model Training (Hours 1–6)

- Download CASIA 2.0 and set up EfficientNet-B4 training pipeline
- Implement ELA map generation and train CNN-ELA combined model
- Load pre-trained ViT-B/16 and CLIP-ResNet50 for AI-image detection
- Implement AutoGAN-style spectrum classifier for GAN artifact detection
- Set up EasyOCR pipeline with YOLOv8 text region detection

Phase 2 — Integration & API (Hours 6–14)

- Build FastAPI backend with complaint image intake endpoint
- Implement Celery parallel task queue for concurrent module execution
- Build Authenticity Confidence Score aggregation engine
- Integrate SHAP + Grad-CAM for explainability report generation
- Set up PostgreSQL schema for complaint tracking

Phase 3 — Dashboard & Custom Dataset (Hours 14–20)

- Build React.js admin dashboard with real-time score visualization
- Photograph and prepare 200+ product images for custom dataset
- Create manipulated versions (expiry date, label edits) and label dataset
- Fine-tune OCR pipeline on custom product packaging images
- Implement heatmap overlay visualization on complaint images

Phase 4 — Demo Preparation (Hours 20–24)

- End-to-end testing: submit genuine images vs manipulated images and verify scores
- Prepare live demo scenario: expiry date manipulation detection in real time
- Performance benchmark: accuracy, precision, recall, F1, latency
- Record backup demo video in case of live demo issues
- Prepare pitch and system walkthrough for judges

12. Model Evaluation Metrics & Targets

Metric	Target	Importance	Context
CNN Accuracy	> 93%	Forgery detection	Baseline: ResNet-101 at 93.46% on CASIA 2.0 (IEEE 2024). VeriSight targets 95%+
ViT AI-Image Accuracy	> 95%	Synthetic detection	ViTs achieve up to 96% accuracy on AI-generated image detection tasks
CLIP GAN Accuracy	> 99%	GAN fingerprinting	CLIP-ResNet50: 99.4% on StyleGAN, 99.2% on StyleGAN2 (CVPR 2023)
OCR Date Extraction F1	> 88%	Text verification	R-CNN achieves 91.1% precision, 80.9% F1 on pharmaceutical packaging (SCPE 2024)
Overall System Precision	> 91%	Minimize false flags	High precision protects genuine customers from unjust refund rejections
Overall System Recall	> 89%	Catch max fraud	High recall ensures manipulated images are not approved as genuine
End-to-End Latency	< 5 seconds	Real-time usability	Parallel module execution via Celery targets sub-5 second total response time

13. Unique & Differentiating Features of VeriSight

Feature	Description
Expiry Date Fraud Focus	VeriSight is the only system that specifically targets digitally altered expiry dates on product packaging — the most common manipulation vector in quick commerce and grocery delivery fraud
Four-Layer Parallel Analysis	CNN forensics + Transformer AI detection + GAN artifact analysis + OCR verification all run simultaneously — no single point of failure; each layer catches different fraud types that others might miss
Platform-Agnostic API	REST API design means VeriSight can be integrated into any delivery, e-commerce, or quick commerce platform in under a day — Zomato, Swiggy, Blinkit, BigBasket, Amazon Fresh, Flipkart
Visual Heatmap Evidence	Generates Grad-CAM + GCA-Net heatmap overlays on complaint images showing exactly which regions were identified as manipulated — makes decisions auditable and legally defensible
Explainable Score Report	Every Authenticity Confidence Score comes with a SHAP-based breakdown explaining which of the four modules flagged the image and why — human reviewers can understand and override every decision
Focused on Deployment, Not Demo	Architecture choices (FastAPI, Docker, ONNX, REST APIs) are all production-deployment decisions — VeriSight is designed to be actually integrated into real systems, not just shown in a hackathon demo
OCR as Supporting Evidence	OCR-extracted text plausibility analysis is used to support (not replace) image forensics decisions — this nuanced design prevents OCR failures from causing wrong decisions

14. Why Should You Select VeriSight?

Real-World Problem at Massive Scale

Returns fraud cost merchants USD 103 billion in 2023. In India, 45% of consumers have encountered deepfake shopping scams. This is not a hypothetical future problem — it is happening right now, at scale, on every platform your judges use.

Addresses a Gap No One Has Solved

No automated system specifically verifies the authenticity of user-submitted product complaint images. Every platform relies on human reviewers. VeriSight solves a real, current, unaddressed gap in the industry's fraud defense stack.

Grounded in Rigorous Research

22+ peer-reviewed papers from IEEE, Springer, NeurIPS, CVPR, MDPI, and ACM directly support each technical decision in VeriSight. This is not AI buzzword stacking — every module has a research-validated foundation.

Practical, Integrable, and Deployable

VeriSight is built with production deployment as the primary design constraint. REST APIs, Docker containers, ONNX inference, async processing — every technical choice reflects a system that is meant to actually be used, not just demoed.

Explainable, Auditable, and Fair

VeriSight does not black-box fraud decisions. Every Authenticity Confidence Score comes with a full explanation. Genuine customers who are incorrectly flagged can be reviewed transparently. This is ethical AI design in practice.

15. Elevator Pitch

> *Every day, delivery platforms blindly trust customer-uploaded images to approve refunds — and fraudsters know it. They alter expiry dates, manipulate labels, and use AI tools to generate fake complaint photos that look completely real. VeriSight stops them. Our AI-powered visual verification system analyzes every complaint image across four layers — image forensics, AI-generation detection, GAN artifact analysis, and OCR text verification — producing an Authenticity Confidence Score in under 5 seconds. No manual reviewer needed for the clear cases. Every decision is explainable. Every manipulation is visualized. VeriSight doesn't just detect fraud — it makes platforms smarter, fairer, and more trustworthy for everyone.*

— End of VeriSight Comprehensive Project Document —
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